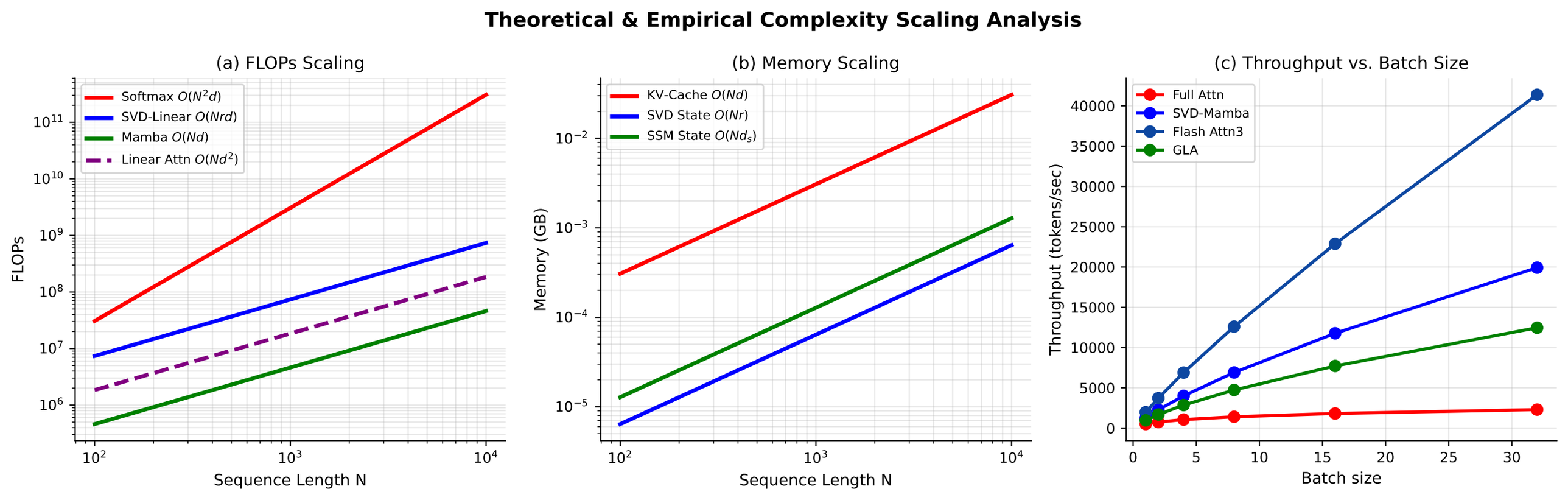
Theoretical & Empirical Complexity Scaling Analysis
(a) FLOPs Scaling
Softmax $O(N^2d)$
SVD-Linear $O(Nrd)$
Mamba $O(Nd)$
Linear Attn $O(Nd^2)$
FLOPs
Sequence Length N
(b) Memory Scaling
KV-Cache $O(Nd)$
SVD State $O(Nr)$
SSM State $O(Nd_s)$
Memory (GB)
Sequence Length N
(c) Throughput vs. Batch Size
Full Attn
SVD-Mamba
Flash Attn3
GLA
Throughput (tokens/sec)
Batch size